

ED + MCMC Direction Reconstruction

3 Categories, 120 Events

PERFORMANCE SUMMARY

Cosine of Angle (True vs Reconstructed Direction)

Mean: -0.2685
Median (p50): -0.4404
68th percentile: -0.0433 ← PRIMARY METRIC
90th percentile: 0.5596
Std deviation: 0.5244

Angular Error (degrees)

Mean: 107.56°
Median: 116.13°
68th percentile: 131.26°
90th percentile: 143.83°

INTERPRETATION

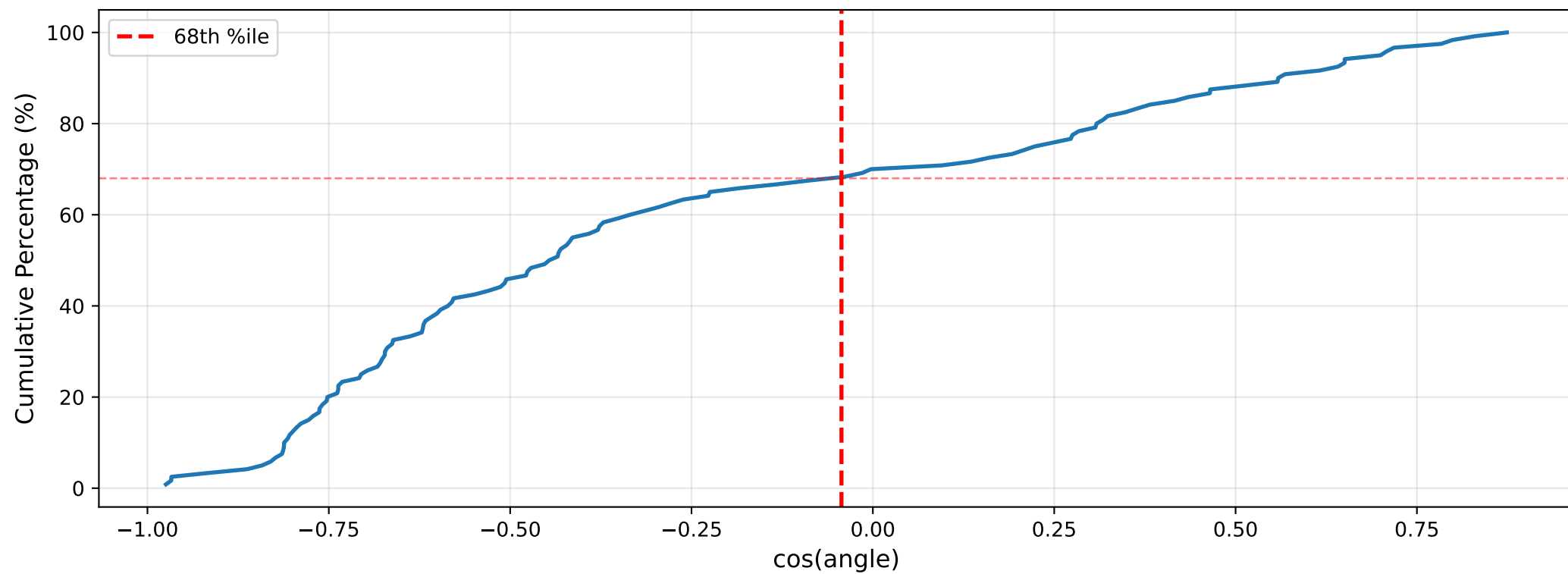
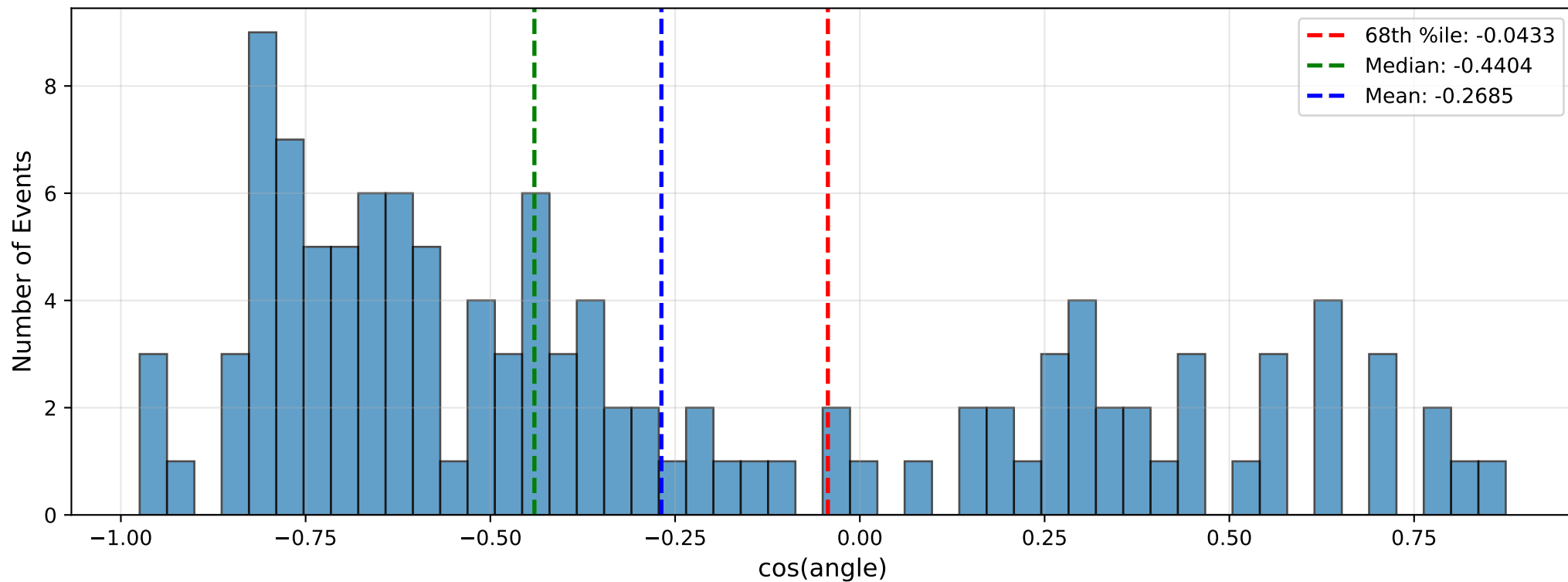
68% of events have $\cos(\text{angle}) \geq -0.0433$
→ 68% of events within 92.48° angular resolution

Perfect pointing: $\cos(\text{angle}) = 1.0$ (0°)

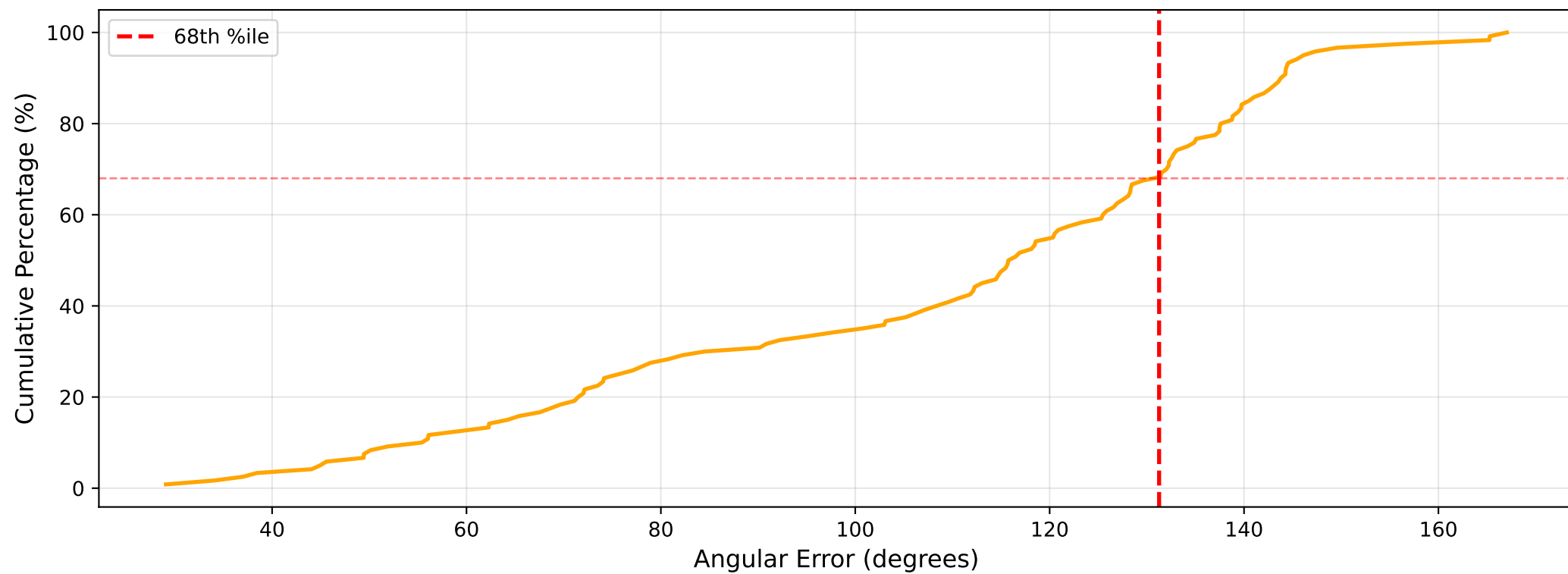
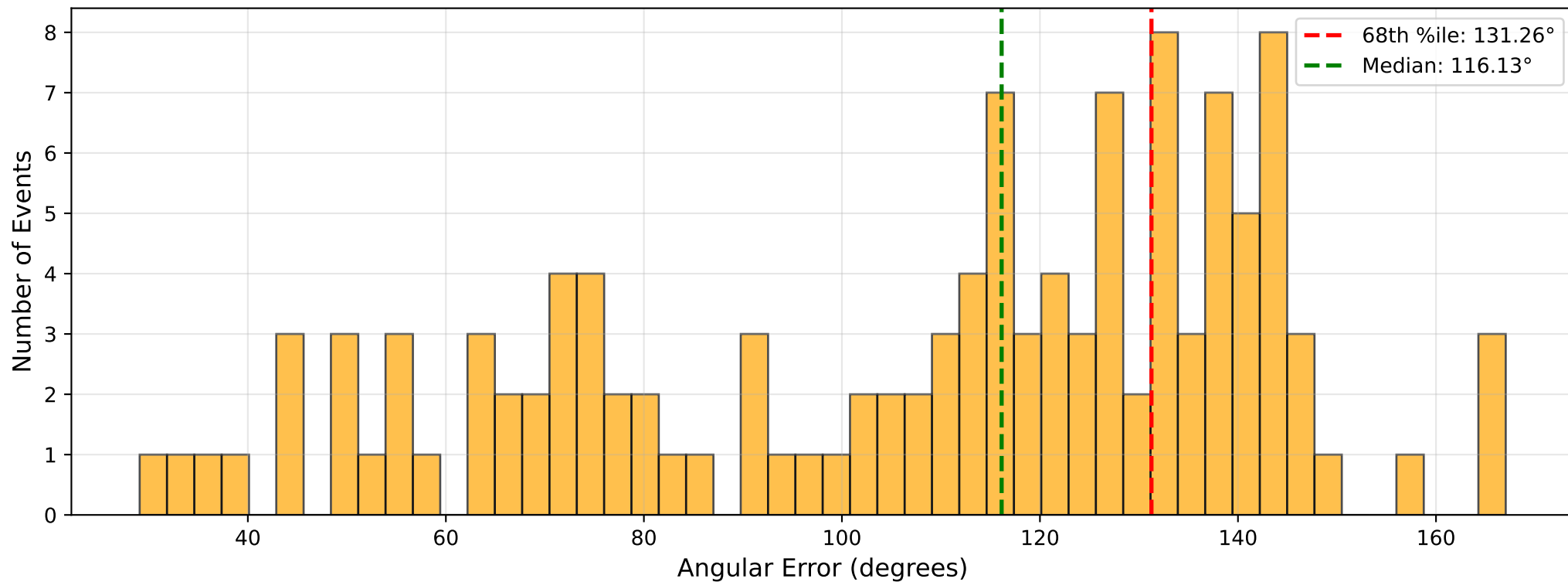
Random pointing: $\cos(\text{angle}) \sim 0.0$ (90°)

Dataset: 3 categories × ~40 events/cat

Distribution of $\cos(\text{angle})$ Between True and Reconstructed



Angular Error Distribution (degrees)



Performance by Category

